Hierarchy induction, one placement decision per step

Toy film-domain vocabulary, routed through the current two-call algorithm (parent check, then child check) with a capacity-triggered consolidation call between types.

- **PARENT CHECK** — does focal belong under a candidate?
- **CHILD CHECK** — does focal subsume a candidate?
- **CONSOLIDATE** — fires between types once a level exceeds B
- **MERGE** — same_concept flag on a parent-check match

(a) Priority bands

SETUP

1. audiovisual work
2. documentary film
3. film
4. movie
5. war film, action film, biographical film

ranked by Weeds-precision in-degree (τ, band tolerance δ)

⚠ **documentary film outranks film here**
in-degree is a frequency proxy, not true generality
— a realistic mis-tiering risk the LLM must correct later.

(b) Step 1 — audiovisual work

```
audiovisual work ← new root
```

roots = {} → no candidates to compare against

settles as a new root

band 1 always runs first and sequentially

(c) Step 2 — documentary film

```
audiovisual work
   ↳ documentary film
```

PARENT CHECK vs {audiovisual work}
→ match, not same concept → **descend**

no further candidates here → **settle**

too shallow — film doesn't exist yet (fixed next)

(d) Step 3 — film

REPARENT

```
audiovisual work
   ↳ documentary film
```

parent of → reparent

```
↳ film
   ↳ documentary film
```

PARENT CHECK vs {audiovisual work} → match → descend
PARENT CHECK vs {documentary film} → no match
CHILD CHECK vs {documentary film} → match → **REPARENT**
a later, more general node corrects an earlier too-shallow attachment

(e) Step 4 — movie

MERGE

```
audiovisual work
   ↳ film ⊕movie
      ↳ documentary film
```

PARENT CHECK vs {audiovisual work} → descend
PARENT CHECK vs {film} → same_concept=true → **MERGE**

“movie” folds in as a surface form — no new node created

(f) Step 5 — war film

```
audiovisual work
   ↳ film ⊕movie
      ├ documentary film
      └ war film
```

PARENT / **CHILD CHECK** vs {documentary film}
→ none / none → **settles as a plain sibling**

film now has 2 children at this level

(g) Step 6 — action film

```
audiovisual work
   ↳ film ⊕movie
      ├ documentary film
      ├ war film
      └ action film
```

PARENT / **CHILD CHECK** vs {doc. film, war film}
→ none / none → settles as a 3rd sibling

no shared parent is invented here —
same_class is not a per-pair action anymore

(h) Step 7 — consolidation

CONSOLIDATE

biographical film descends to film’s level:
3 candidates > B (batch cap) → **CONSOLIDATE** fires first

```
↳ film
   ├ documentary film
   └ genre film (new)
      ├ war film
      └ action film
```

LLM groups TRUE siblings only (same kind of thing);
documentary film stays out of the group.

a capacity response, not a per-pair judgment —
never fires below B candidates

(i) Step 8 — biographical film

FINAL TREE

PARENT CHECK vs {doc. film, genre film}
→ match (documentary film) → descend → settle

```
audiovisual work
   ↳ film ⊕movie
      ├ documentary film
      │  └ biographical film
      └ genre film
         ├ war film
         └ action film
```

final tree after all eight steps

Every merge, reparent, and consolidation decision is adjudicated by a dedicated LLM call — never inferred from embedding similarity alone.